
Python Chapter: List, Tuple, Dictionary – Deep Notes with Code & Differences

❑ What are Lists, Tuples & Dictionaries?

In Python, these are **collection data types** used to store **multiple items** in a single variable. Har ek ka apna kaam, style aur use-case hota hai.

1. List – (Changeable & Ordered)

✓ Notes:

- List ek **ordered, changeable (mutable)** collection hai.
- Items ko **index number** se access karte hain.
- List me **duplicate values** allowed hain.

❑ Syntax:

```
my_list = [10, 20, 30, "hello", True]
```

Common List Methods:

Method	Use	Example
<code>append(x)</code>	List ke end me value add karta hai	<code>my_list.append(99)</code>
<code>insert(i, x)</code>	i-th index pe item insert karta hai	<code>my_list.insert(2, 50)</code>
<code>remove(x)</code>	Specific item ko remove karta hai	<code>my_list.remove(20)</code>
<code>pop(i)</code>	i-th index ka item hata deta hai	<code>my_list.pop(0)</code>
<code>sort()</code>	List ko ascending me sort karta hai	<code>my_list.sort()</code>
<code>reverse()</code>	List ko ulta karta hai	<code>my_list.reverse()</code>
<code>clear()</code>	Saare items hata deta hai	<code>my_list.clear()</code>

```

my_list = [10, 20, 30, "hello", True]
my_list.append(99)
print(my_list) # Output: [10, 20, 30, 'hello', True, 99]

my_list.insert(2, 50)
print(my_list) # Output: [10, 20, 50, 30, 'hello', True, 99]

my_list.remove(20)
print(my_list) # Output: [10, 50, 30, 'hello', True, 99]

my_list.pop(0)
print(my_list) # Output: [50, 30, 'hello', True, 99]

nums = [5, 3, 8, 1]
nums.sort()
print(nums) # Output: [1, 3, 5, 8]

nums.reverse()
print(nums) # Output: [8, 5, 3, 1]

nums.clear()
print(nums) # Output: []

```

🔍 Example:

```

fruits = ["apple", "banana", "cherry"]
fruits.append("mango")
print(fruits) # Output: ['apple', 'banana', 'cherry', 'mango']

```

🌀 2. Tuple – (Fixed & Ordered)

✓ Notes:

- Tuple ek **ordered, unchangeable (immutable)** collection hota hai.
- **Faster** than lists (speed me tez hota hai).
- Duplicate values allowed hoti hain.

☐ Syntax:

```
my_tuple = (10, 20, 30, "hello", True)
```

⚠ Agar ek hi item ka tuple banana hai: `single_item = (5,)` (comma zaroori hai)

☐ Tuple Me Kya Kar Sakte Ho?

- Access: `my_tuple[1]`
- Count: `my_tuple.count(10)`
- Index: `my_tuple.index("hello")`
- Convert to list: `list(my_tuple)` (taaki modify kar sako)

🔍 Example:

```
numbers = (1, 2, 3, 4)
print(numbers[2])      # Output: 3
print(numbers.count(2)) # Output: 1
```

□ 3. Dictionary – (Key-Value Based)

✓ Notes:

- Dictionary ek **unordered, changeable, key-value** based data structure hai.
- Har value ek unique key ke sath store hoti hai.
- Duplicate keys allowed **nahi** hain.

□ Syntax:

```
my_dict = {
    "name": "Ravi",
    "age": 21,
    "is_student": True
}
```

📁 Common Dictionary Methods:

Method	Use	Example
<code>get(key)</code>	Key ki value deta hai	<code>my_dict.get("name")</code>
<code>keys()</code>	Saare keys deta hai	<code>my_dict.keys()</code>
<code>values()</code>	Saari values deta hai	<code>my_dict.values()</code>
<code>items()</code>	Key-value pairs deta hai	<code>my_dict.items()</code>
<code>update({...})</code>	Dictionary me new data add/update karta hai	<code>my_dict.update({"age": 22})</code>
<code>pop(key)</code>	Specific key-value ko hata deta hai	<code>my_dict.pop("age")</code>

🔍 Example:

Example with Code and Output:

```
my_dict = {
    "name": "Ravi",
    "age": 21,
    "is_student": True
}

print(my_dict.get("name"))      # Output: Ravi
print(my_dict.keys())          # Output: dict_keys(['name',
print(my_dict.values())        # Output: dict_values(['Ravi'
print(my_dict.items())         # Output: dict_items([('name',

my_dict.update({"age": 22})
print(my_dict)                 # Output: {'name': 'Ravi', 'age': 22}

my_dict.pop("age")
print(my_dict)                 # Output: {'name': 'Ravi', 'is_student': True}
```

```
person = {"name": "Amit", "age": 25}
print(person["name"])          # Output: Amit
person["city"] = "Delhi"
print(person)                  # Output: {'name': 'Amit', 'age': 25, 'city': 'Delhi'}
```

Difference Table: List vs Tuple vs Dictionary

Feature	List	Tuple	Dictionary
Syntax	[]	()	{ key: value }
Ordered?	✓ Yes	✓ Yes	✗ No (Python 3.7+ me mostly yes)
Mutable?	✓ Yes	✗ No	✓ Yes
Duplicates?	✓ Allowed	✓ Allowed	✗ Keys not allowed duplicate
Access method	Index	Index	Key
Use Case	Changeable data	Fixed set of items	Store related info with label

```
# List is mutable
my_list = [1, 2, 3]
my_list[0] = 100
print(my_list) # Output: [100, 2, 3]

# Tuple is immutable
my_tuple = (1, 2, 3)
# my_tuple[0] = 100 ✗ Error: 'tuple' object does not support item assignment

# Dictionary is mutable
my_dict = {"name": "Ravi"}
my_dict["name"] = "Rahul"
print(my_dict) # Output: {'name': 'Rahul'}
```